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|  | Work Instruction                                        | Doc. Revision  | 00         |
|                                                                                  | AXI V810 X Axis Kollmorgen Module Removal & Replacement | Effective Date | 2 Oct 2016 |

This procedure is applicable to below models:

|           |                                     |    |                                     |    |
|-----------|-------------------------------------|----|-------------------------------------|----|
| V810 STD  | <input checked="" type="checkbox"/> | S1 | <input checked="" type="checkbox"/> | S2 |
| V810 XXL  | <input checked="" type="checkbox"/> | S1 | <input checked="" type="checkbox"/> | S2 |
| V810 S2EX | <input checked="" type="checkbox"/> |    |                                     |    |
| V810 Mini | <input checked="" type="checkbox"/> |    |                                     |    |

#### REVISION HISTORY

| Rev | Effective Date | Description   | Doc Originator |
|-----|----------------|---------------|----------------|
| 00  | 2 Oct 2016     | First Release | ONG KEAN SHENG |
|     |                |               |                |
|     |                |               |                |

|                                                                                  |                                                                    |                |                   |
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## X Axis Kollmorgen Removal

1. Unload board from machine if any
2. Close X ray Source at Operator Console
3. Close SSCA
4. Identify X axis Kollmorgen (PN,4XASP-A170)location by refer below Figure 1.

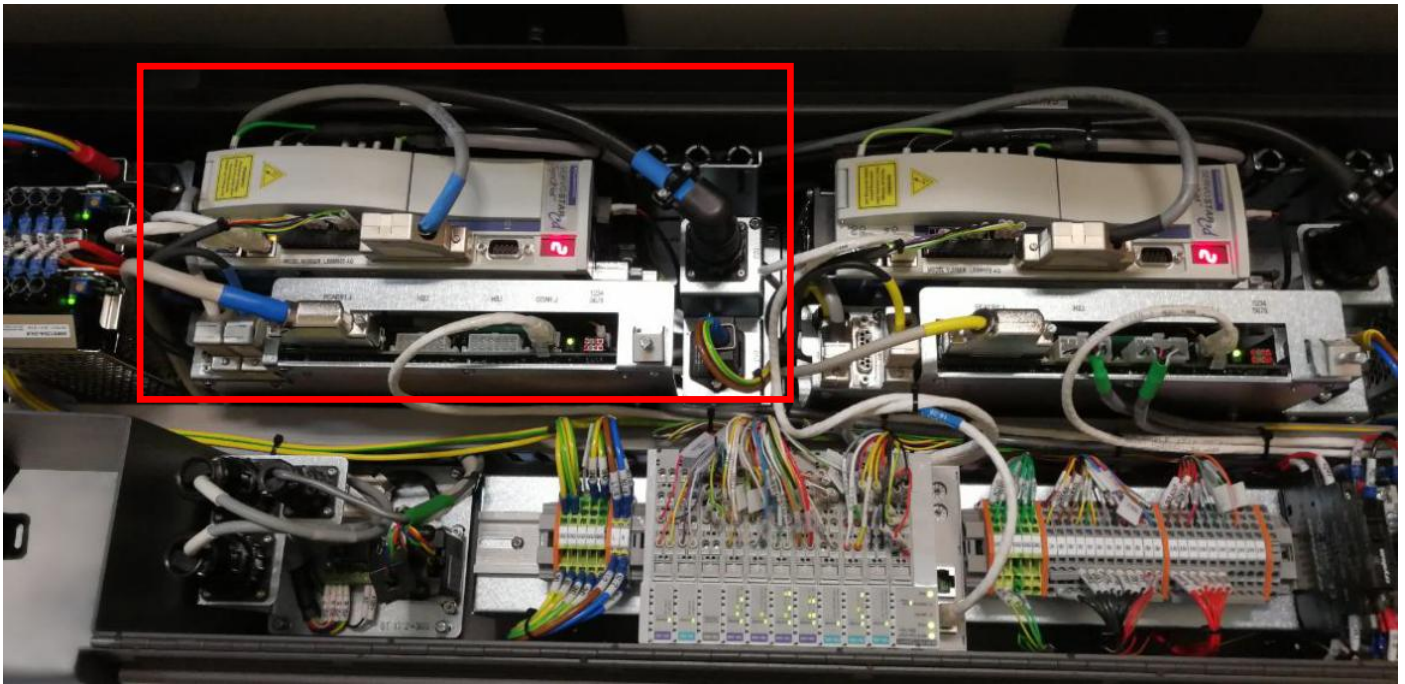


Figure 1

5. Disconnect Power Connector from 01J as below Figure 2.

Remarks :

If Kollmorgen is equipped with Vertical Type Power Connector, Retaining Clip will require to remove in advance

If Kollmorgen is equipped with Horizontal Type Power Connector, may unplug directly

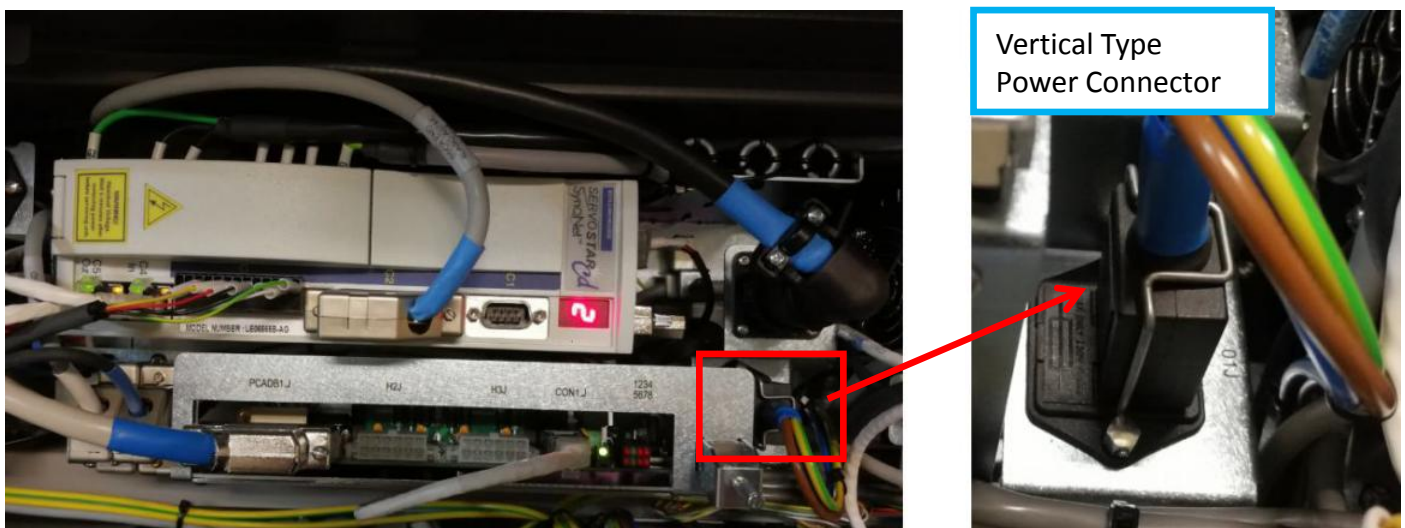


Figure 2



|                                                                                  |                                                         |                |            |
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6. Disconnect X-Axis Motor Power Cable as Figure 3.

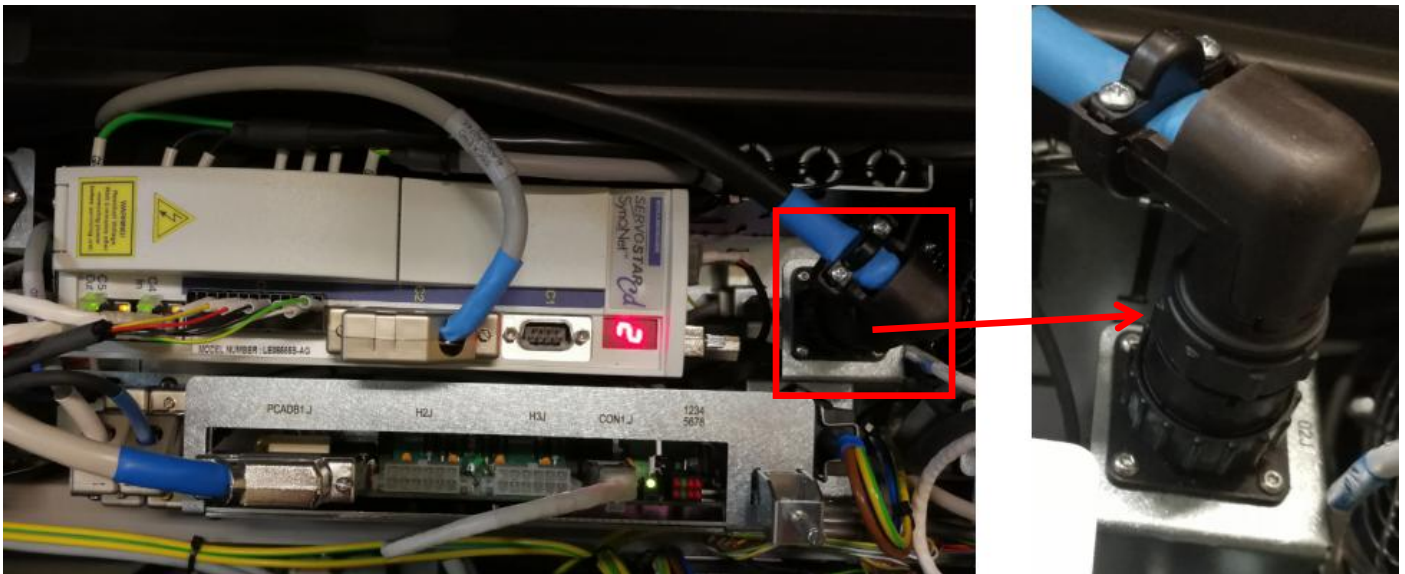


Figure 3

7. Disconnect X-Axis Motor Rotary Cable as Figure 4

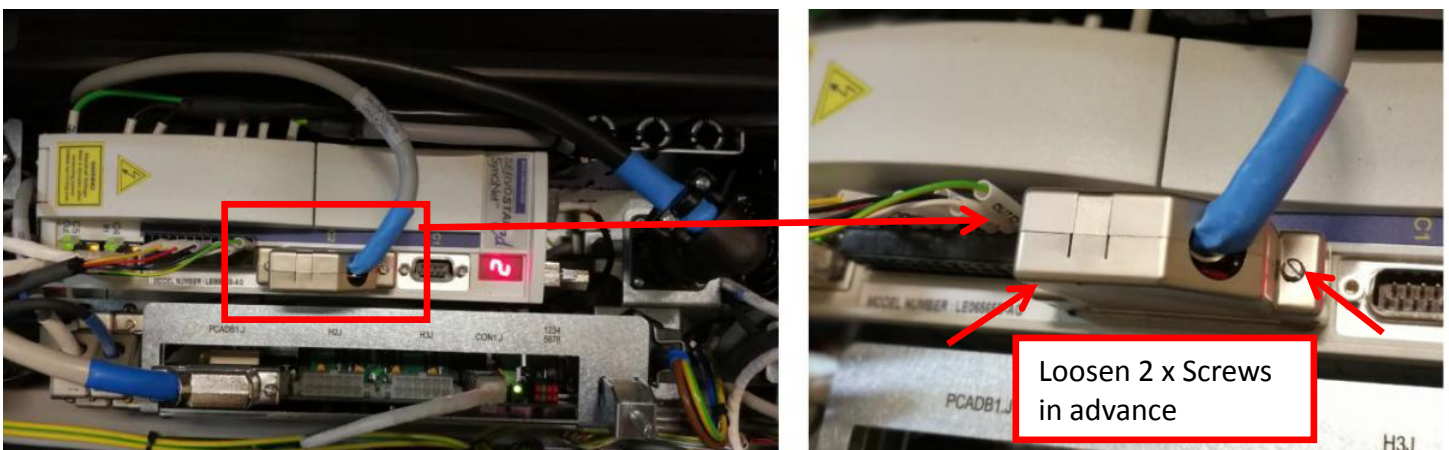


Figure 4

8. Disconnect 2 x Cat 5 Cable from C5 Out & C4 In as below Figure 5.

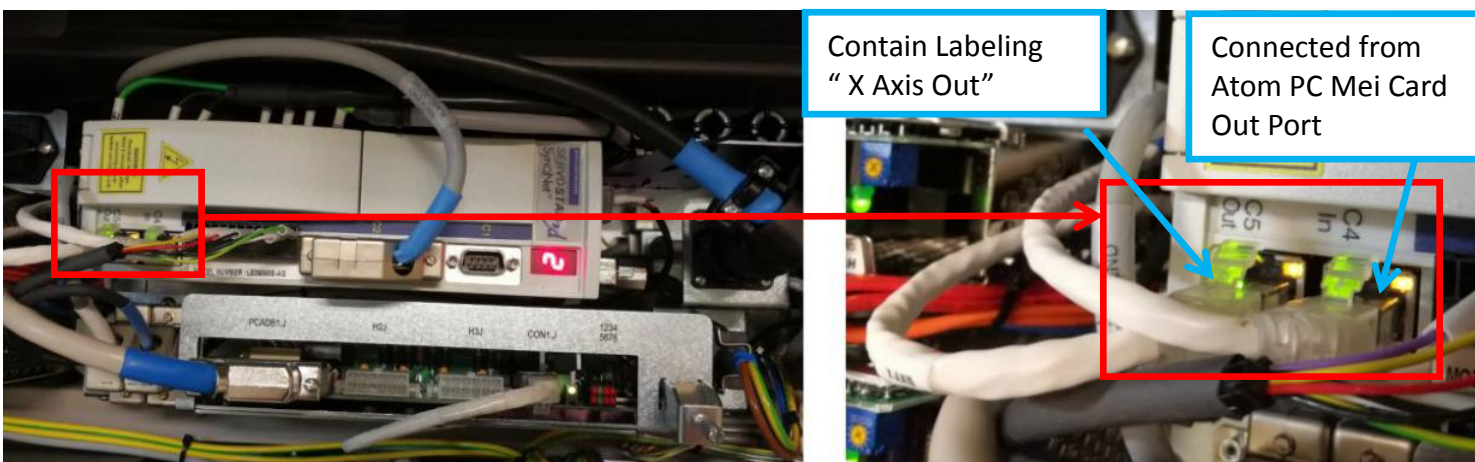


Figure 5

|                                                                                  |                                                         |                |            |
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9. Disconnect 2 x Connector (24VDC & 5 VDC) from 03J & 04J as Figure 6.

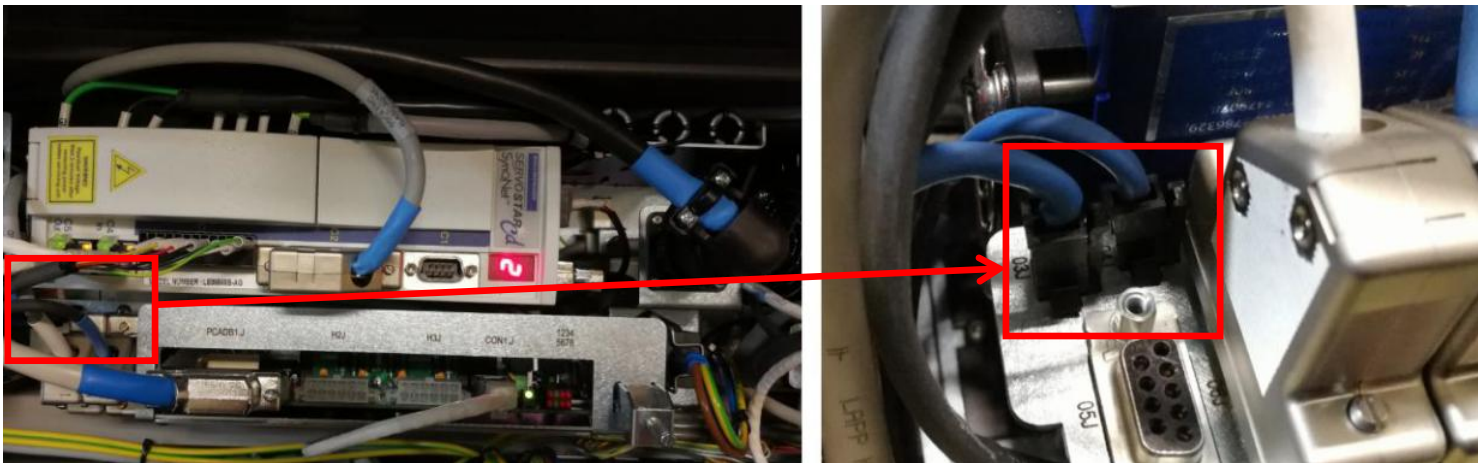


Figure 6

10. Disconnect X axis encoder & AWA Home Sensor connector from 07J & 06J as Figure 7

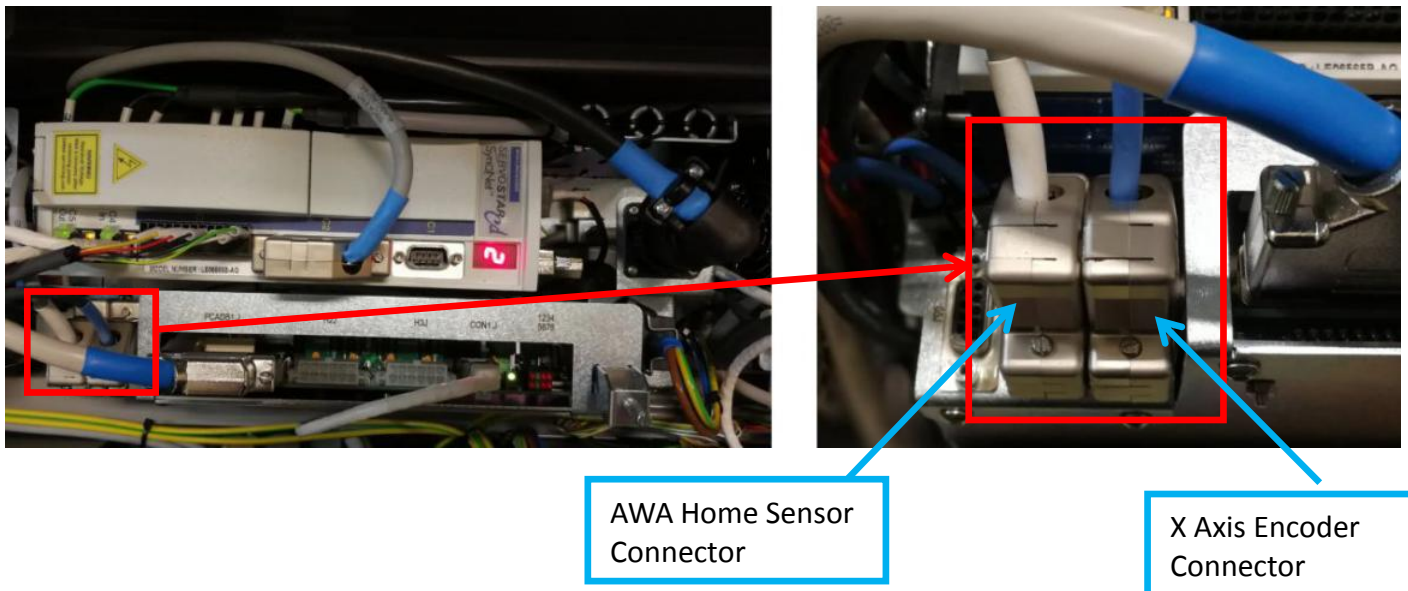


Figure 7

11. Disconnect Stepper Motor Driver connector from PCAADB1J as Figure 8

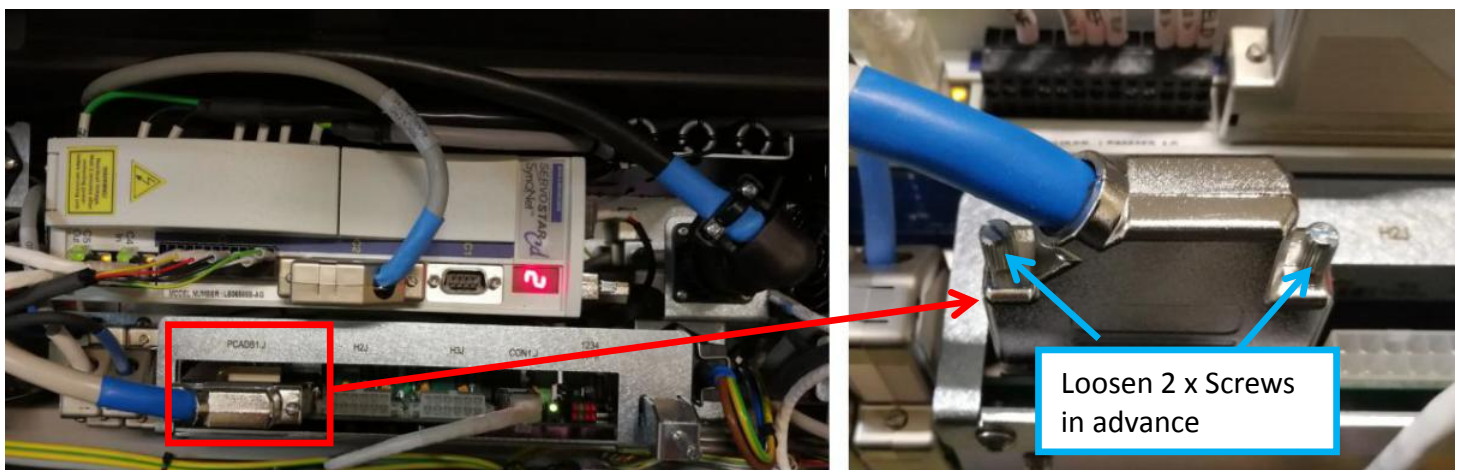


Figure 8



|                                                                                  |                                                         |                |            |
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12. Disconnect X axis Interlock Cat5 Cable from CON1J as Figure 9.

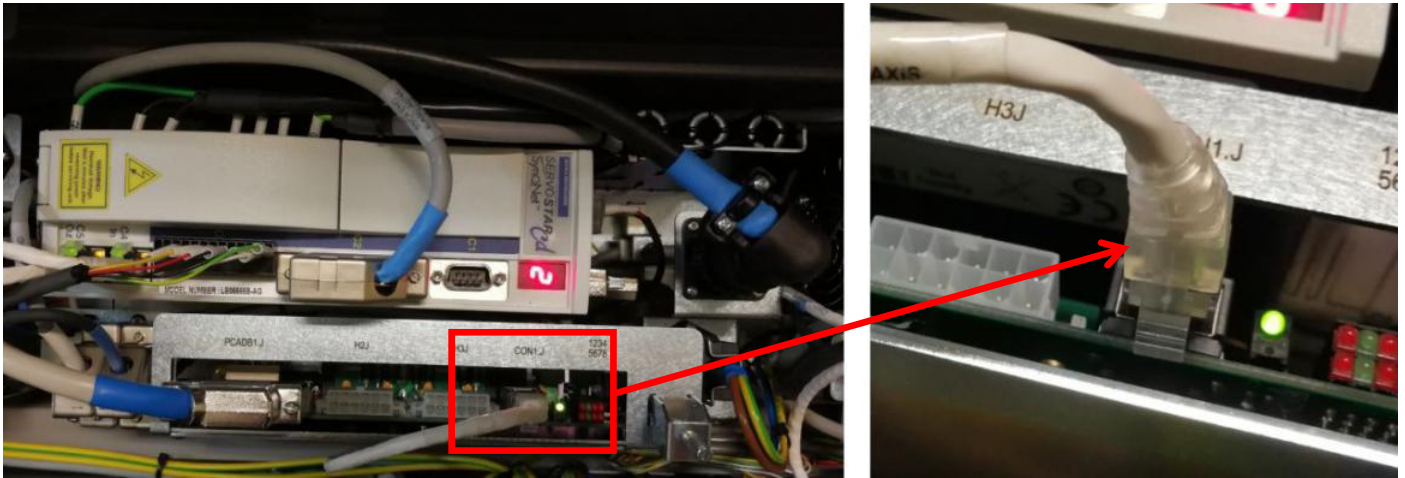


Figure 9

13. Loosen 4 x Screws at Kollmorgen Module Support base as Figure 10.



Figure 10

14. Gently remove entire X Axis Kollmorgen module from SSCA

## X Axis Kollmorgen Replacement

15. May refer to below table for replacement step

| Model          | Machine SN             | Referring Step |
|----------------|------------------------|----------------|
| V810 Std S1/S2 | MY10133156 & BELOW     | Step 16~41     |
| V810 XXL S1/S2 | MY13138020 & BELOW     | Step 16~41     |
| V810 S2EX      | MY10133156S2EX & BELOW | Step 16~41     |
| V810 Std S1/S2 | MY10133157 & ONWARD    | Step 27~41     |
| V810 XXL S1/S2 | MY13138021 & ONWARD    | Step 27~41     |
| V810 S2EX      | MY10133157 & ONWARD    | Step 27~41     |

16. Identify Horizontal Type Power Connector where remove at Step 5. Loosen Philip screws at Horizontal Type of Power Connector as Figure 11

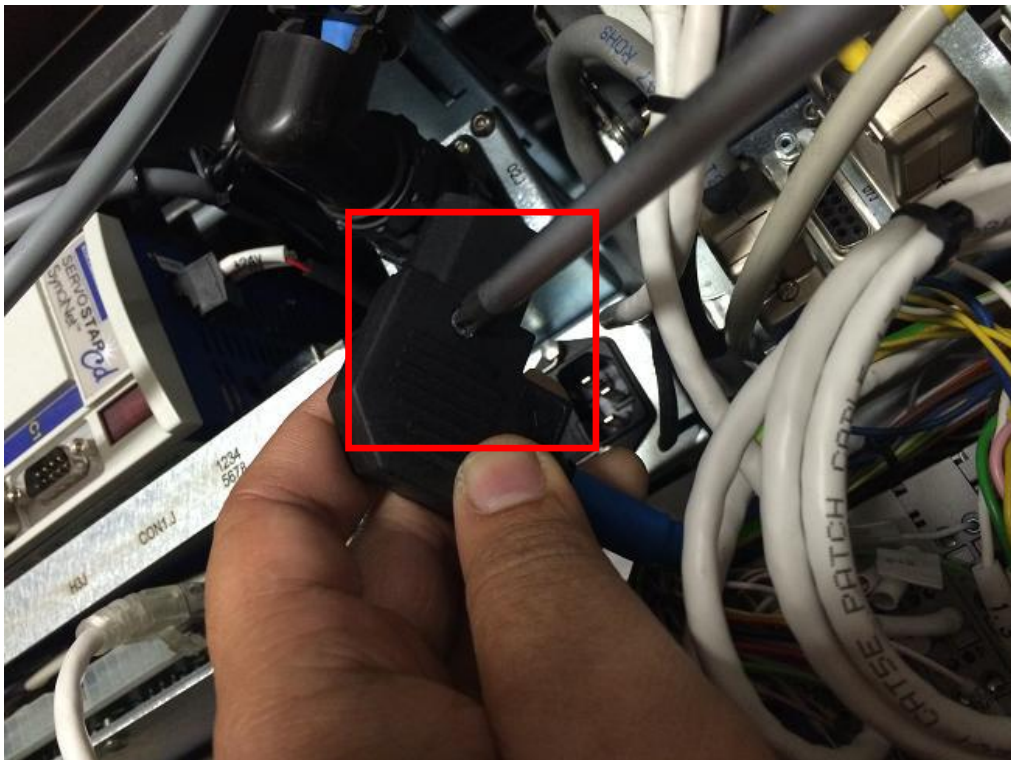


Figure 11

|                                                                                                |                                                                    |                |                   |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------|-------------------|
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17. Open up insulator cover & identify Life(L),Ground(G) & Neutral(N) as Figure 12.

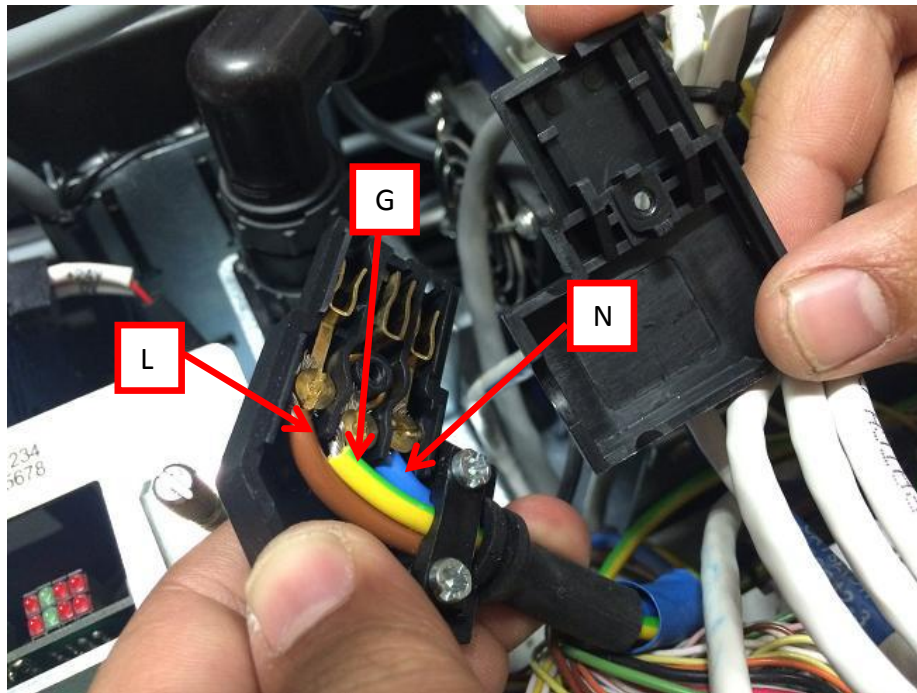


Figure 12

18. Loosen 3 x Flat Head Screw Terminal as Figure 13



Figure 13



|                                                                                  |                                                         |                |            |
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19. Loosen 2 x Philip screws and remove cable clamp as Figure 14

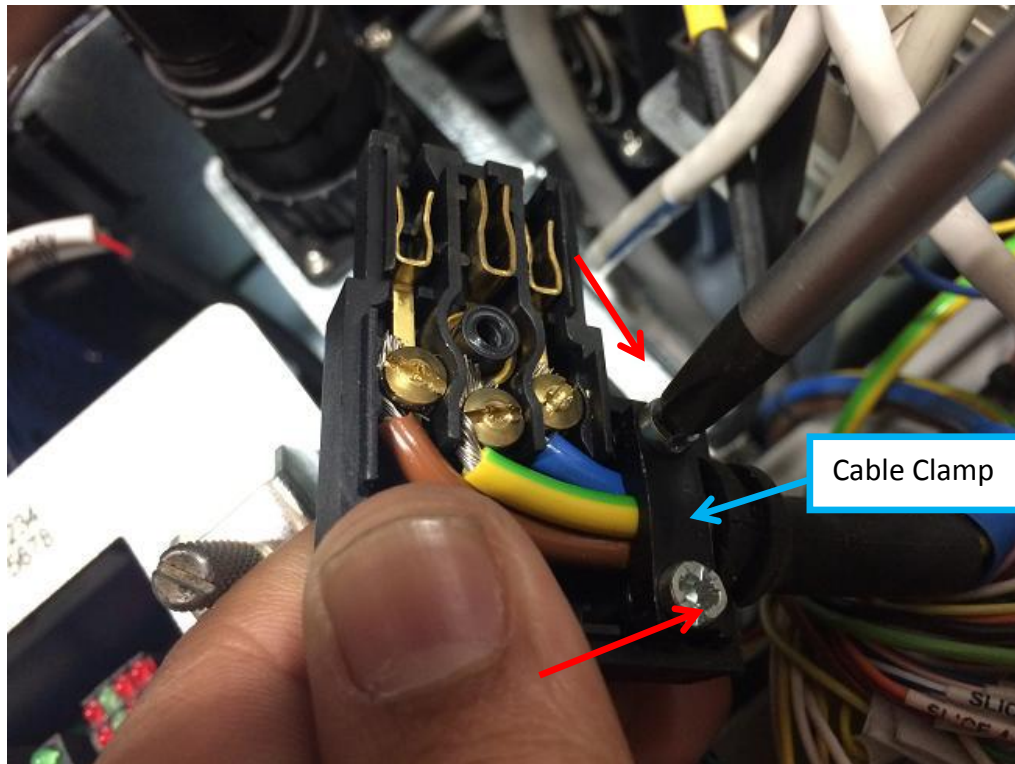


Figure 14

20. Remove 3 x Wire Cores from Horizontal Type Power Connector as Figure 15.

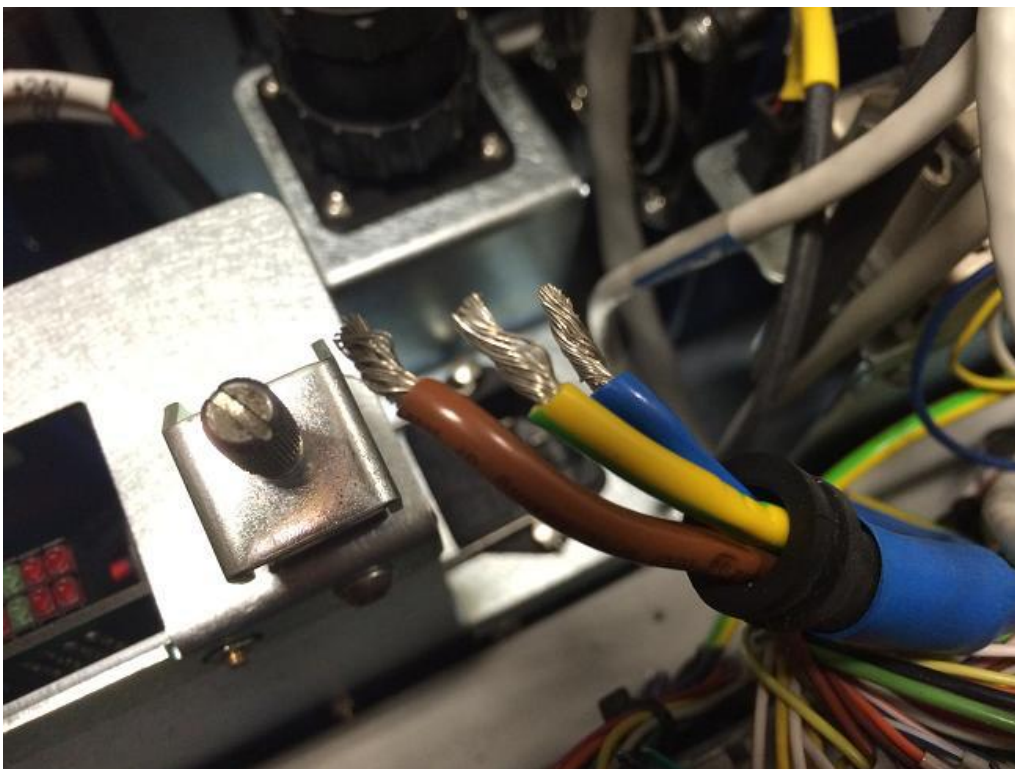


Figure 15

|                                                                                  |                                                         |                |            |
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21. Gently Cut Brown(Life),Green/Yellow(Ground) & Blue(Neutral) into equivalent length. Strip away cable jacket by Cable Stripper.

Remarks : Caution on cable cutting length to avoid insufficient cable length

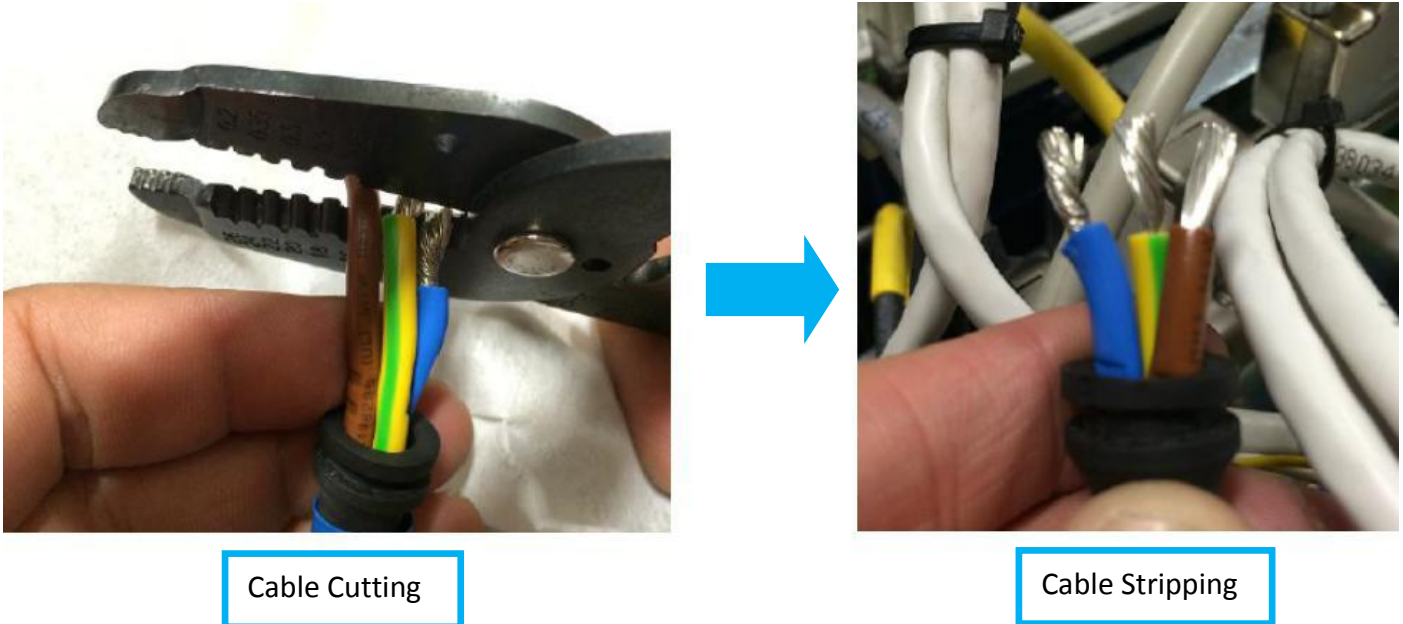


Figure 16

22. Unplug Vertical Type Power Connector from Kollmorgen Module as below Figure 17.

Remarks : Each Kollmorgen Spare Module (4XASP-A170) will equipped with 1 x vertical connector.

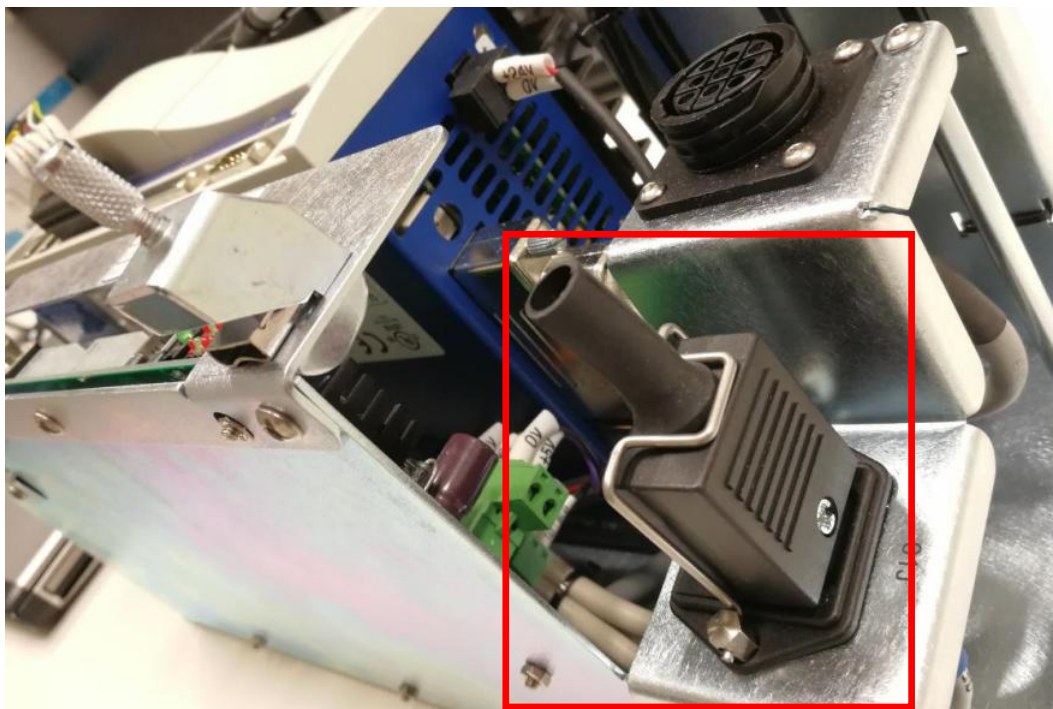


Figure 17



|                                                                                  |                                                         |                |            |
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23. Open Insulator Cover from Vertical Type Power Connector as below Figure 18. Loosen 2 x Screws on the cable clamp & remove accordingly.

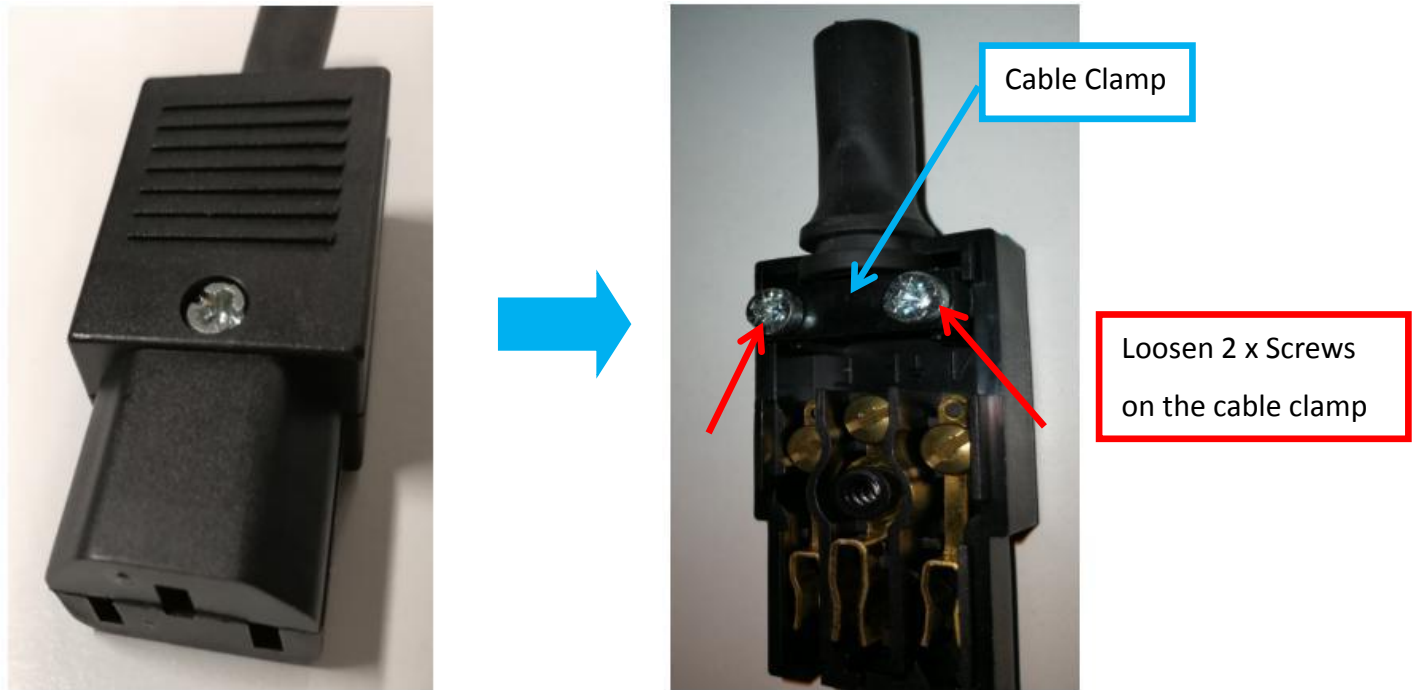


Figure 18

24. Identify the terminal for Life(L), Ground(G) & Neutral & assemble all three cables into respective screw terminals. May refer to below Figure 19.

Remarks : Sleeve Ring which include with Vertical Type Power Connector will not require for assembly.

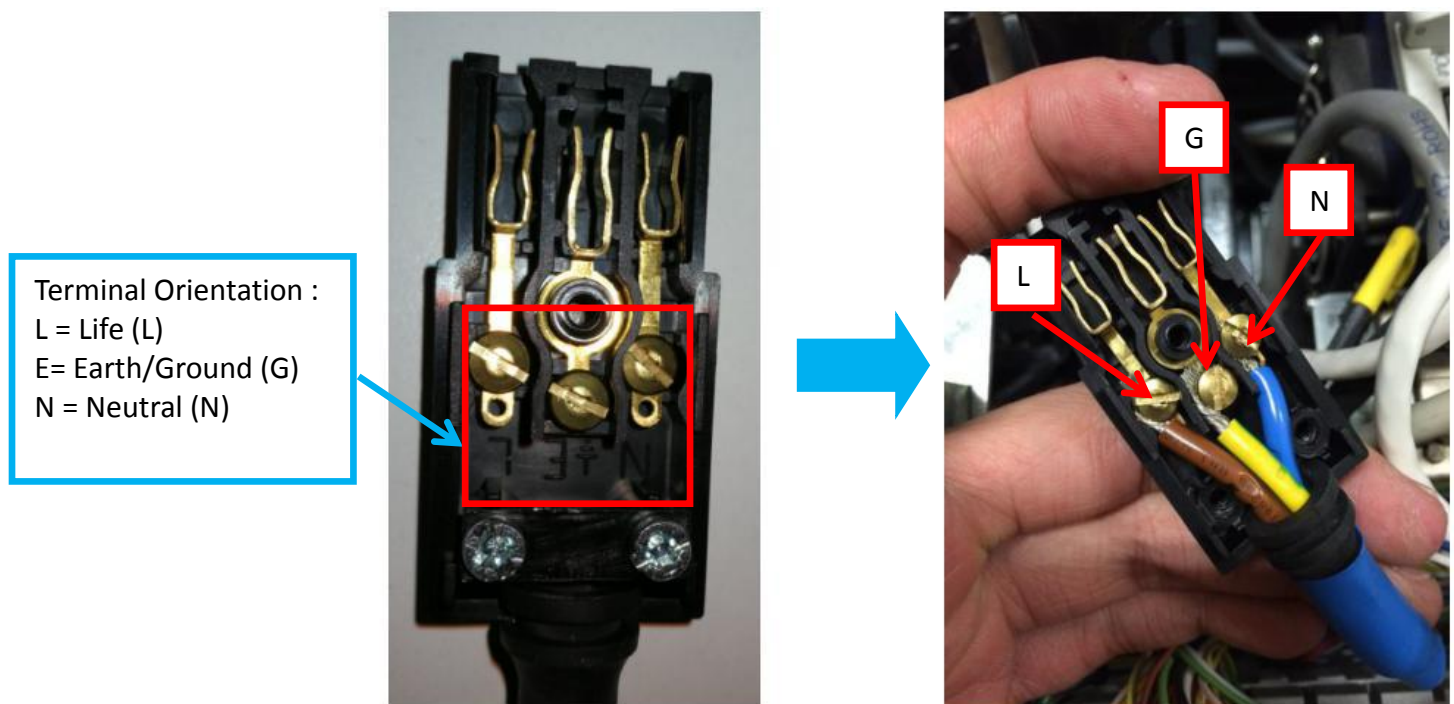
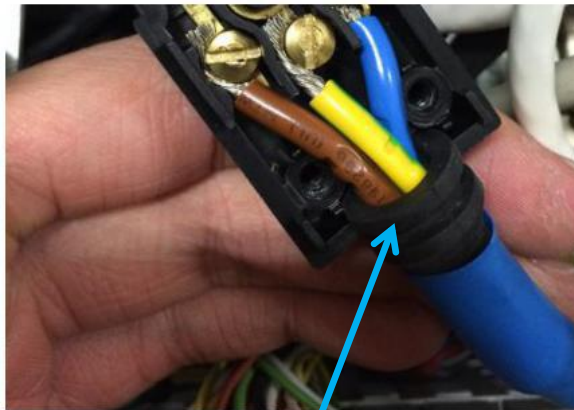


Figure 19

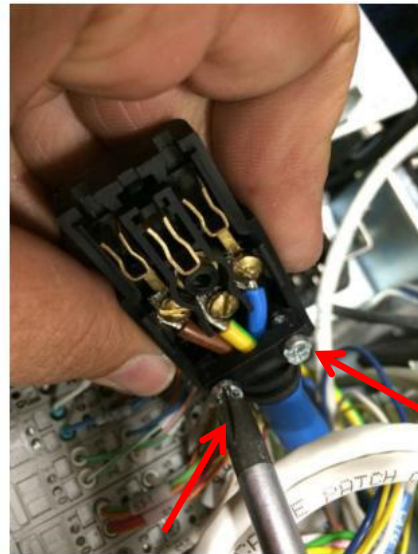


|                                                                                  |                                                         |                |            |
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25. Gently position sleeve ring where attach with existing power cable into slot as Figure 20. Assemble cable clamp by tighten 2 x Philip screws as Figure 20.



Sleeve ring sit firmly in the slot



Tighten 2 x Philip screws

Figure 20

26. Assemble the Insulator Cover at Vertical Type Power Connector as Figure 21.

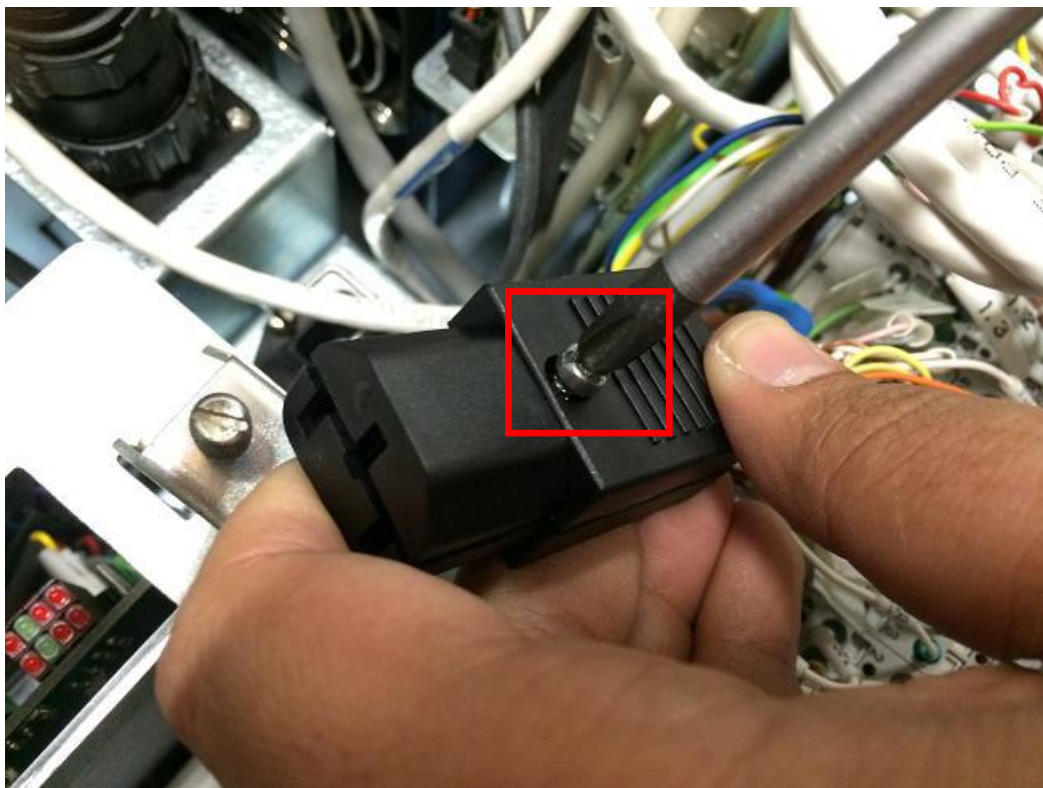


Figure 21

|                                                                                  |                                                         |                |            |
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27. Gently slot X-Axis Kollmorgen (Pn, 4XASP-A170) into SSCA. Tighten 4 x Screws at Kollmorgen Module Support base as Figure 22.



Figure 22

28. Connect X axis Interlock Cat5 Cable at CON1J as Figure 23 .

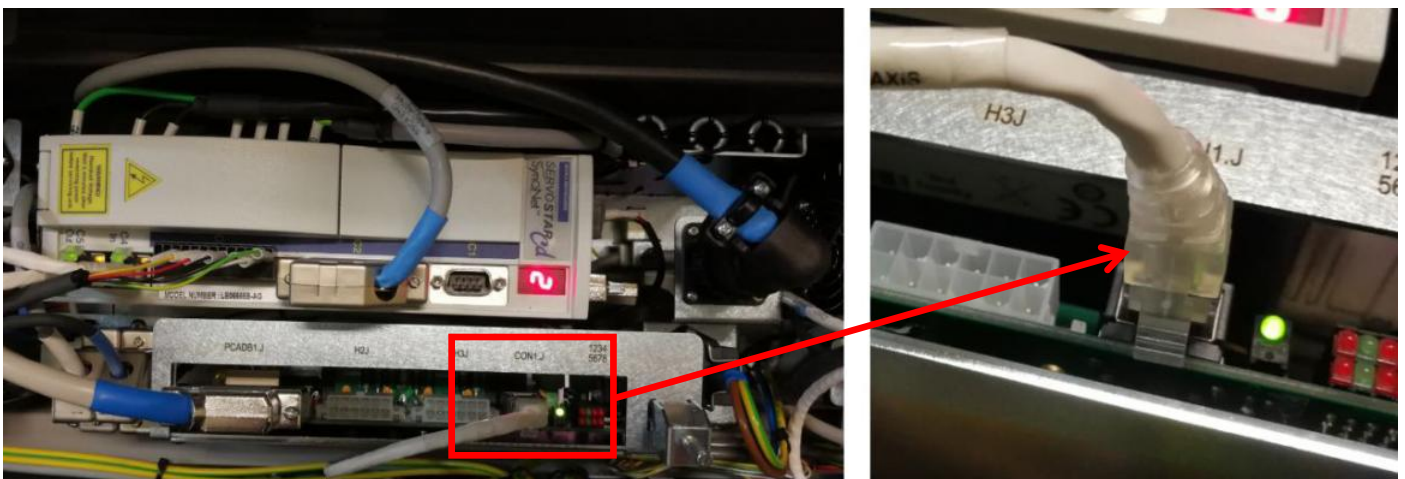


Figure 23

29. Connect Stepper Motor Driver connector at PCAADB1J as Figure 24

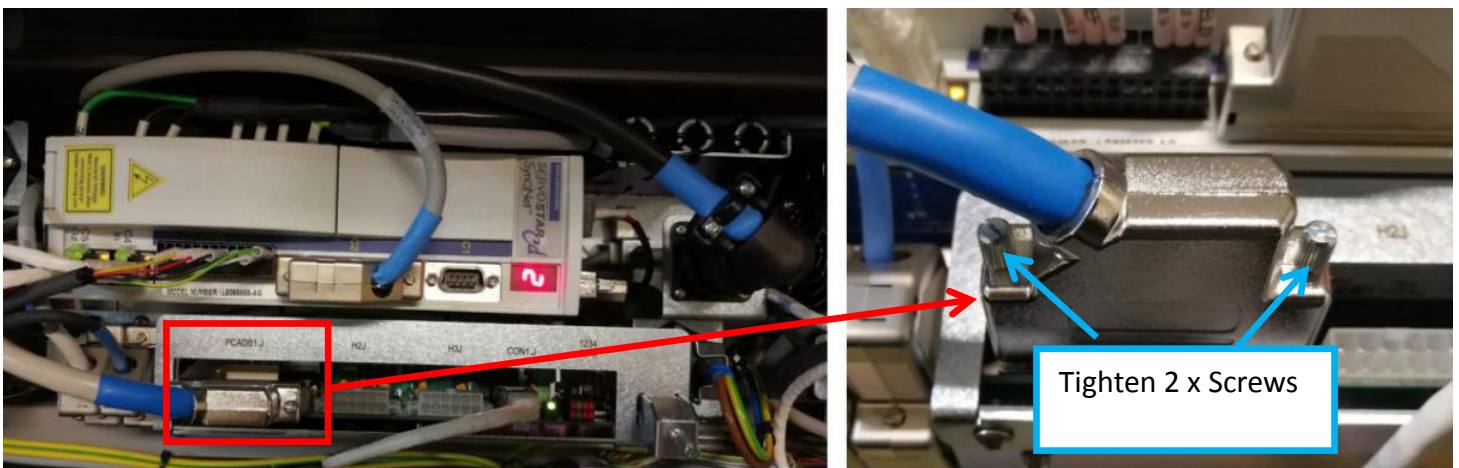


Figure 24



|                                                                                  |                                                         |                |            |
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30. Connect X axis encoder & AWA Home Sensor connector at 07J & 06J as Figure 25

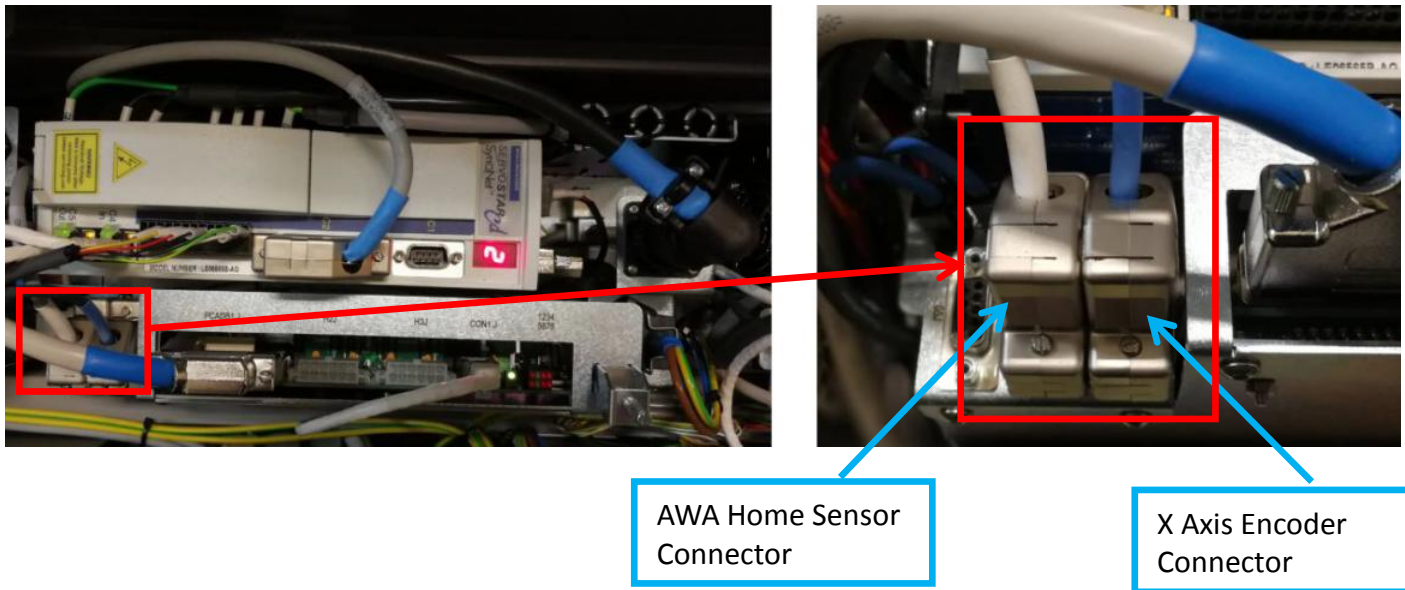


Figure 25

31. Connect 2 x Connector (24VDC & 5 VDC) at 03J & 04J as Figure 26.

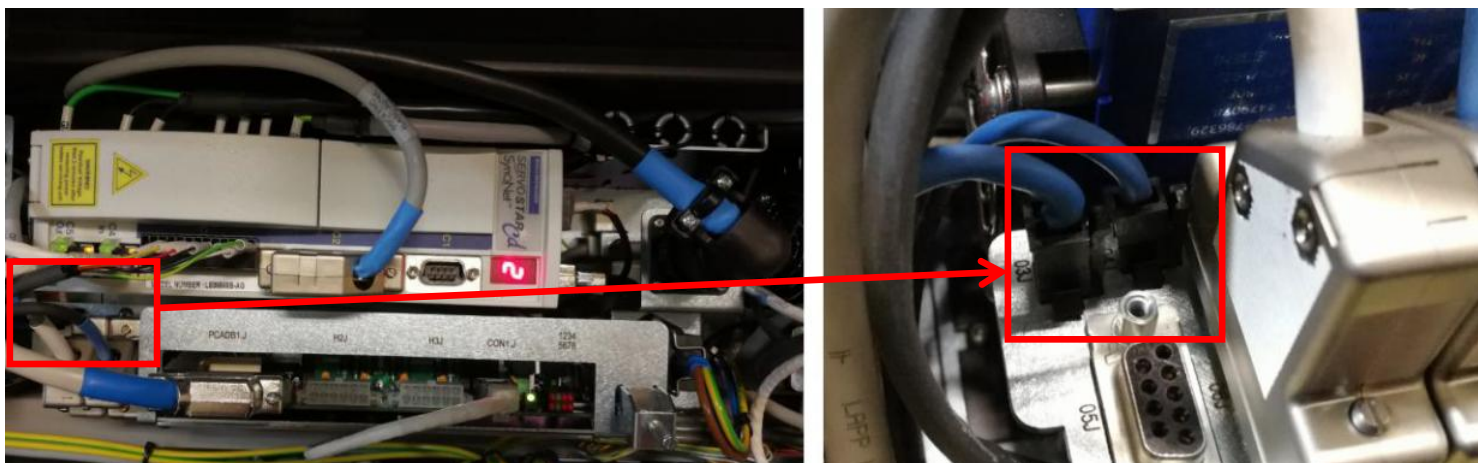


Figure 26

32. Connect 2 x Cat 5 Cable from C5 Out & C4 In as below Figure 27.



Figure 27



|                                                                                  |                                                         |                |            |
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### 33. Connect X-Axis Motor Rotary Cable as Figure 28

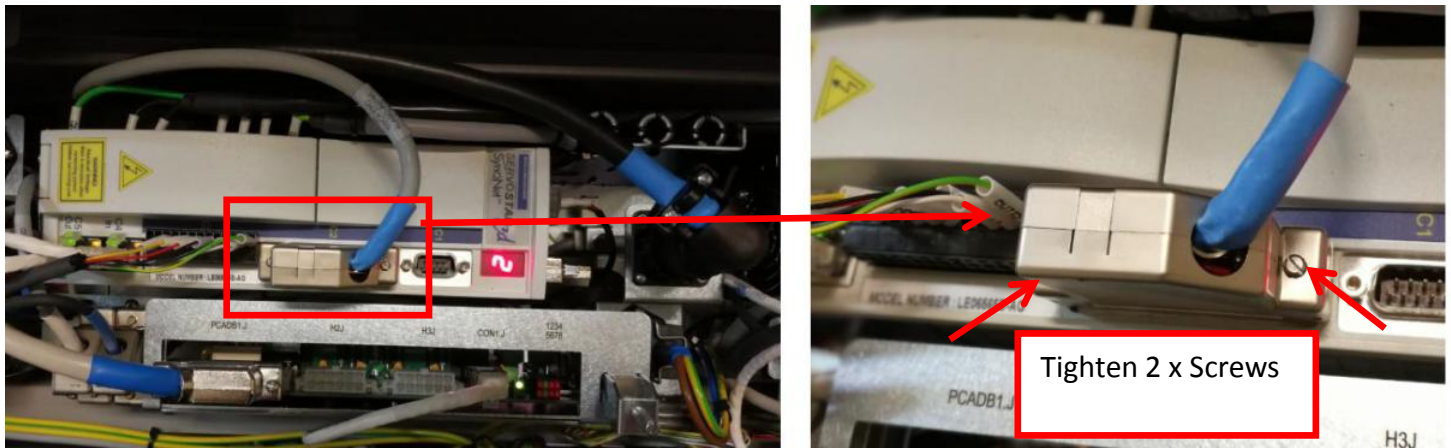


Figure 28

### 34. Connect X-Axis Motor Power Cable as Figure 29.

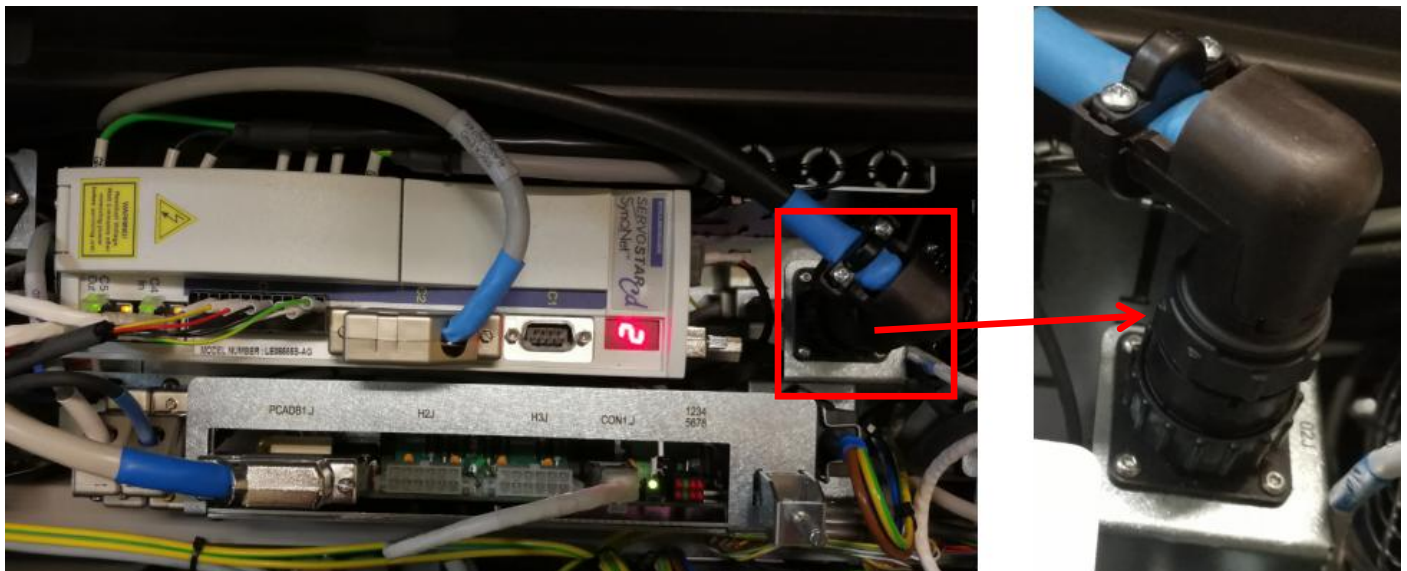


Figure 29

### 35. Connect Power Connector at 01J as below Figure 30.

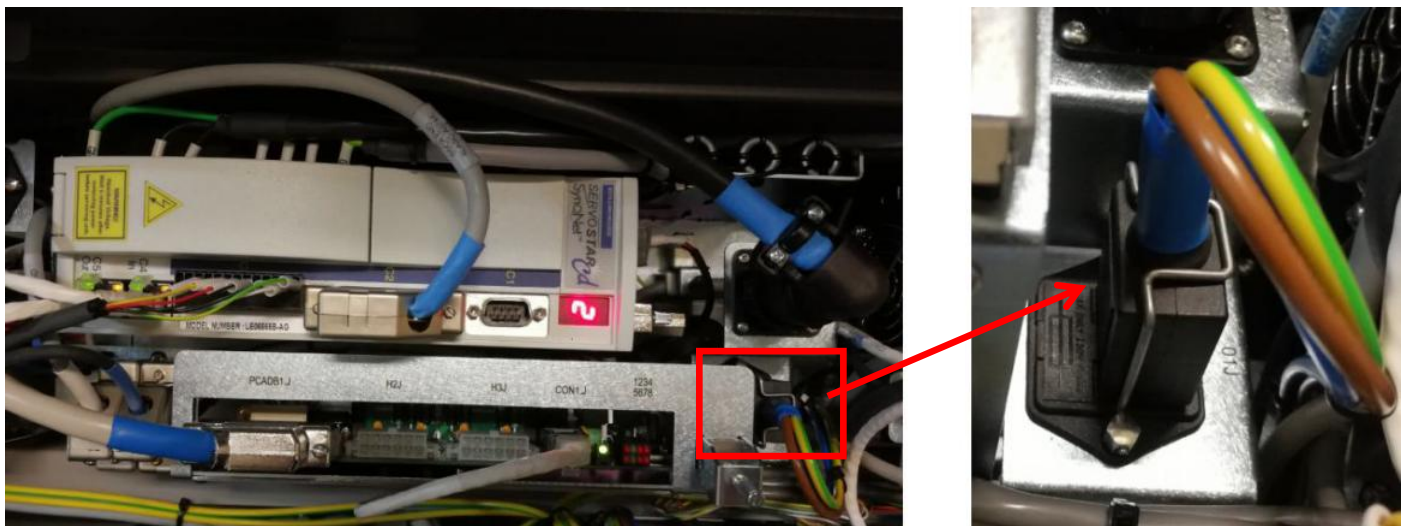


Figure 30

|                                                                                  |                                                                    |                |                   |
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36. Turn On SSCA
37. Intialize at Digital IO Page
38. Perform “ Upgrade Firmware” at Motion Control tab
39. Intialize at Motion Control tab
40. Turn on X ray Source at Operator Console Panel
41. Run 1 Loop of CD & A.